The scalar glueball on center-vortex-removed SU(2) lattice ensembles: a data note

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Abstract

We measure the zero-momentum 0^{++} glueball correlator on SU(2) Wilson-action ensembles at $\beta = 2.3$, 2.4 and 2.5 in three readings of the same configurations: the original links, the center-projected Z_2 links in direct maximal center gauge, and the vortex-removed links of de Forcrand and D'Elia. A literature search found no published glueball measurement on vortex-removed ensembles; this is new as far as we found, not proven new. The cleanest point in this data set is $\beta = 2.4$, with the caveat that only two couplings enter any vortex-removed mass comparison and there is no continuum statement. There, the vortex-removed string tension from our short-time estimator is zero within errors; a reanalysis of the stored Wilson loops done after referee review finds the long-time estimators consistent with zero only at the largest distances we resolve (Creutz ratios at $R = 6, 7$), with a residual rise at intermediate distances ($\chi(5,5)$ is 0.12(2) of the original). The vortex-removed 0^{++} effective masses are consistent with a constant over $t = 1$ to 3, with mass $2.40(8)\sqrt{\sigma}$ in our own string-tension unit ($2.52(8)\sqrt{\sigma}$ in the literature unit), 0.71(4) of the original glueball mass, and unchanged between 12^4 and 16^4 (ratio 0.98(4), 5.6σ from the box ratio 0.75). The $\beta = 2.4$ vortex-removed mass is from one gauge copy and carries an untested gauge-copy systematic; at the neighbouring couplings the best of 16 copies moved the mass by about 20%. At $\beta = 2.5$ the vortex-removed mass is also volume independent ($20^4/16^4$ ratio 0.94(6)), but our pre-registered premise check fails there: the short-time estimator leaves about 0.21 of the original string tension on the vortex-removed ensemble, and so do E1′ ($R \geq 4$) and $\chi(6,6)$; E2 at $T = 5, 6$ is at or below zero and $\chi(7,7)$ is consistent with zero. With the best of 16 gauge copies the residual does not fall (0.225(10)), the center-projected string tension drops to 0.73 of the full one, and the vortex-removed mass shifts by 2.3σ. At $\beta = 2.3$ the vortex-removed effective mass does not plateau on the larger box. Applying our pre-registered rules exactly, the combined outcome is inconclusive. We publish the measurements, the failure modes, data and code.

1. Limitations, stated first

- Monte Carlo evidence on small lattices: boxes from 8^4 to 20^4, at most two volumes per coupling in each volume test.
- SU(2) only, Wilson action, three couplings ($\beta = 2.3, 2.4, 2.5$), no continuum extrapolation. Only 2.4 and 2.5 enter any vortex-removed mass comparison, because the $\beta = 2.3$ channel is not under control (sec. 4.2). Nothing here bears on the continuum mass-gap problem.
- The vortex-removed (VR) ensemble is gauge dependent: it is defined through direct maximal center gauge (MCG), which has many local maxima (Gribov copies). The main runs use one gauge copy per configuration. The gauge-copy study (sec. 4.4) was done at $\beta = 2.5$ and at a $\beta = 2.3$ control, not at $\beta = 2.4$; there the VR mass moved with the copy by $+0.109(48)$ and $-0.192(109)$ respectively, about 20% of the mass. Copies closer to the global maximum (simulated annealing,

many copies [14]) and Laplacian center gauge [2] were not tested; the VR mass may depend on that choice.

- VR configurations are not generated by a local reflection-positive action, so VR correlators need not be positive sums of exponentials; "mass" means the decay rate read by our estimator.
- Removing the vortices flips the sign of the trace on every P-plaquette, so the VR plaquette is much lower than the original (0.585 vs 0.630 at $\beta = 2.4$, 0.623 vs 0.652 at 2.5). VR configurations therefore carry a density of lattice-scale defects, and the VR 0^{++} channel could be partly a defect (ultraviolet) effect. The only check here is that $m_{\mathrm{VR}}/\sqrt{\sigma}$ changes little between two couplings ($2.52 \to 2.38$ in literature units).
- Estimator caveats: effective masses are resolved only to $t \approx 3$; the volume tests use a fixed plateau rule on $m(1), m(2)$; the pre-registered string-tension premise check uses $V(R) = \ln W(R,3)/W(R,4)$, which probes short physical times and, with a free Coulomb term, can absorb part of a potential that rises and then flattens inside the fit window (sec. 4.3). With this estimator our units $a\sqrt{\sigma}$ differ from the literature values [9, 29] by factors 1.04 ($\beta = 2.3$, 16^4), 0.95 (2.3, $16^3 \times 32$), 1.05 (2.4) and 1.10 (2.5); we give masses in both units. In literature units the original glueball masses are 1 to 6% below the continuum value $3.78(7)\sqrt{\sigma}$ [10].
- Polyakov loops were not measured on the VR ensembles, so we cannot say whether center symmetry is broken there at $T = L$, which bears on which finite-volume states the 0^{++} operators can couple to (sec. 6).
- Not blind: the original-ensemble glueball mass was known in advance; the series 1 estimators were chosen after seeing exploratory 8^4 and 12^4 data, and those exploratory data are part of the series 1 sample (sec. 3); one deviation (GEVP t_0) was made after testing on already-seen data; the long-time estimators of sec. 4.3 and the gauge-copy method of sec. 4.4 were chosen after the $\beta = 2.5$ premise failure was known; the $\beta = 2.4$ long-time reanalysis was done after referee review. The rules of each series were written and committed to a private git repository before that series' new production data existed (internally time-stamped, not externally registered).

2. Background: what is known and what is new

Center vortices as the confining configurations are reviewed in [17]. De Forcrand and D'Elia [1] removed center vortices from SU(2) configurations by multiplying each link by its center projection in direct MCG, and found that confinement is lost (Creutz ratios tend to zero, large-T slopes do not grow with R) at $\beta = 2.4$ on 16^4. Pepe and de Forcrand [2] found the same at $\beta = 2.3$ to 2.5 with Laplacian center gauge. Direct MCG and center dominance go back to Del Debbio et al. [3, 4]. Vortex removal has since been used for gluon and ghost propagators [5, 6, 7], the SU(2) quark propagator [30], the SU(3) string tension [31], and the light hadron spectrum and chiral symmetry in SU(3) [8], where it restores chiral symmetry. Biddle, Kamleh and Leinweber [16] found that the vortex-removed static potential is not described by $V_0 + \sigma R - e/R$ at long range and fitted it with a modified Coulomb term. Glueballs after Abelian projection were measured in [11, 12]. Langfeld and Schafke [13] measured 0^+ and 2^+ screening masses on thick-vortex ensembles obtained by gauge-invariant cooling, which is a different reduction from the one used here. The dependence of center-projected observables on the MCG gauge copy is documented in [14, 15]: better maxima lower the projected string tension, and the effect weakens on larger volumes.

New here, as far as our search found (arXiv, INSPIRE including the papers citing [1], 1999 to October 2026): (a) the 0^{++} glueball correlator on VR ensembles and on thin Z_2 CP ensembles, with volume tests; (b) a quantification of how a short-time string-tension estimator misreads the VR potential (a residual of about 0.2 at $\beta = 2.5$ that long-time estimators do not show); the phenomenon itself is expected from [1, 2], which used long-distance observables, and from the failure of the standard potential form reported in [16]; (c) a paired gauge-copy study of that residual and of the VR mass.

3. Method

Ensembles. SU(2) Wilson action, heat bath [20, 21] with four over-relaxation [22, 23] steps per compound sweep, hot start, 300 thermalization sweeps. Three series, each with independent streams stopped by a wall clock fixed before any of its data were analysed:

- Series 1: $\beta = 2.3$ on 8^4, 12^4, 16^4 and $\beta = 2.4$ on 12^4, 16^4, $T = L$, a measurement every 2 sweeps. Part of this sample existed before the series 1 rules were written: the $\beta = 2.3$ 8^4 data (5000 measurements) and two of the three 12^4 streams at each coupling (1700 of 3670 measurements at $\beta = 2.3$, 1750 of 3730 at $\beta = 2.4$) come from earlier exploratory streams with the same action and gauge fixing, which had been analysed. New after the rules: one 12^4 stream per coupling and all 16^4 streams (five at 2.3, four at 2.4; three of them were added about 2.4 h after launch with the same deadline, before any data were analysed).
- Series 2: $\beta = 2.3$ on $12^3 \times 32$ and $16^3 \times 32$; $\beta = 2.5$ on 16^4 and 20^4; a measurement every 2 sweeps. All data new after the rules.
- Series 3 (gauge-copy study): $\beta = 2.5$ on 16^4 (8 streams) and 20^4 (2 streams), $\beta = 2.3$ on $12^3 \times 32$ (2 streams, control); a measurement every 10 sweeps. All data new after the rules.

Center projection and vortex removal. Direct MCG maximizes $\sum_{x,\mu} |\mathrm{tr}\, U_\mu(x)|^2$ by local exact maximization with over-relaxation ($\omega = 1.85$) until the relative change is below 10^{-8}. With $Z_\mu(x) = \mathrm{sign}\,\mathrm{tr}\, U_\mu(x)$, the CP links are $Z_\mu(x)$ and the VR links are $Z_\mu(x)U_\mu(x)$ [1]. On every plaquette of every configuration we assert $P_{\mathrm{orig}} = Z_P P_{\mathrm{VR}}$, which fails if the gauge fixing is not a gauge transformation. Series 1 and 2 use one gauge copy (over-relaxation started from the configuration as generated). Series 3 fixes every configuration twice: the same single copy ("single"), and the best functional among 16 candidates, the single copy plus 15 Haar-random gauge transforms each relaxed to convergence ("best of 16"). Both are measured on the same configuration, so differences are paired.

Glueball. Zero-momentum sums of spatial Wilson loops [28] (1×1, 2×2, 3×3, 4×4, 1×2, 2×3, 3×4) at three APE smearing levels [24] ($\alpha = 0.5$; $(0, 2, 8)$ steps at $\beta = 2.3$, $(0, 4, 15)$ at 2.4, $(0, 8, 31)$ at 2.5, scaled with $(a\sqrt{\sigma}\,)^{-2}$): basis A, 21 operators (CP: the 7 shapes unsmeared). Vacuum-subtracted correlators averaged over all time slices, normalized by $\sqrt{\mathrm{diag}\, C(0)}$, ground vector from the generalized eigenvalue problem [25, 26] at $(t_0, t) = (0, 1)$, effective masses $m(t) = \ln C(t)/C(t+1)$. Plateau rule, fixed in advance: if $m(2)$ has relative error below 30% and $|m(1) - m(2)| < 1.5\sigma$ (jackknife of the difference), quote $m(1)$ ("plateau"); if $m(2)$ is that precise but differs, quote $m(2)$ ("late"); otherwise quote $m(1)$ as an "upper bound"; a quoted value with relative error above 30% is "noisy".

String tension. Estimator E1: $V(R) = \ln W(R,3)/W(R,4)$, fitted with $V_0 + \sigma R - e/R$, e free, $R = 2, \ldots, L/2$, with spatially smeared Wilson loops. Series 3 and the reanalyses of sec. 4.3 add E1′ (the same fit on $R \geq 4$), E2 (the same fit on $\ln W(R,T)/W(R,T+1)$ at $T = 5$ and $T = 6$) and Creutz ratios $\chi(R,R)$. Ratios $\sigma_X/\sigma_{\mathrm{ORIG}}$ use the same estimator on the same configurations. Literature values used for units and checks: $a\sqrt{\sigma} = 0.369$ ($\beta = 2.3$), 0.266 (2.4), 0.188 (2.5) [9, Table 10; 29, Table 7].

Errors. Delete-one-block jackknife, block length at least twice the largest integrated autocorrelation time of the plaquette and the 2×2 operator (Madras-Sokal window [27], $c = 6$), 8 to 20 blocks per stream; the bin-size scan of the VR mass errors (4 to 40 blocks per stream) is in data/bin_size_scan.csv. Autocorrelation times of the gauge-fixed observables themselves were not estimated. Thermalization check: first quarter dropped if its plaquette differs from the rest by more than 3σ (one stream in series 1).

Pre-registered tests. Premise C0: $\sigma_{\mathrm{VR}}/\sigma_{\mathrm{ORIG}} < 0.15$ and $\sigma_{\mathrm{CP}}/\sigma_{\mathrm{ORIG}} \in [0.6, 1.4]$ with E1. Volume test at each coupling: $\rho = m(L_{\mathrm{large}})/m(L_{\mathrm{small}})$ in lattice units; “gap” if ρ is within 2σ of 1 and more than 2σ from the box ratio $L_{\mathrm{small}}/L_{\mathrm{large}}$, “gapless” the reverse, otherwise inconclusive; interpreted only if the original-ensemble control reads “gap”.

4. Results

4.1 Statistics and checks

Plaquettes: 0.60228 ($\beta = 2.3$), 0.63001 (2.4), 0.6519 to 0.6520 (2.5). At $\beta = 2.3$ and 2.5 they agree within 3×10^{-4} with [29, Table 7] (0.6025, 0.6520); the same table lists $1 - 0.36352 = 0.6365$ at 2.4, which does not lie between its neighbours and which we could not confirm elsewhere; our 0.63001 does. Integrated autocorrelation times are 0.5 to 1.6 measurements (one short stream 3.0). No assertion fired in any stream. MCG functional: 0.753 (2.3), 0.773 (2.4), 0.791 (2.5, single copy). P-vortex density: 0.090, 0.055, 0.032.

4.2 Masses

Table 1. Basis A, lattice units, quoted mass with plateau flag (pl = plateau). VR effective masses $m(1), m(2), m(3)$ in the last column.

β	lattice	N	ORIG	VR	CP	VR $m(1), m(2), m(3)$
2.3	8^4	5000	1.134(54) pl	0.719(83) late	0.863(24) pl	1.022(41), 0.719(83), 0.226(88)
2.3	12^4	3670	0.859(104) late	0.943(28) pl	0.863(19) pl	0.943(28), 0.944(93), 0.458(121)
2.3	16^4	1228	1.049(79) pl	0.691(103) late	0.847(26) pl	1.018(57), 0.691(103), 0.481(158)

β	**lattice**	**N**	**ORIG**	**VR**	**CP**	**VR** $m(1), m(2), m(3)$
2.3	$12^3 \times 32$	1510	1.216(42) pl	1.052(43) pl	0.880(20) pl	1.052(43), 1.004(126), 0.433(126)
2.3	$16^3 \times 32$	840	1.325(67) pl	0.891(104) late	0.926(26) pl	1.050(42), 0.891(104), 0.508(142)
2.4	12^4	3730	0.873(25) pl	0.680(18) pl	0.609(17) late	0.680(18), 0.688(34), 0.548(56)
2.4	16^4	1230	0.942(47) pl	0.669(22) pl	0.640(28) late	0.669(22), 0.661(51), 0.577(91)
2.5	16^4	920	0.709(36) pl	0.475(18) pl	0.419(19) late	0.475(18), 0.467(32), 0.509(56)
2.5	20^4	560	0.701(36) pl	0.448(21) pl	0.371(22) late	0.448(21), 0.416(34), 0.436(56)

Units $a\sqrt{\sigma_{\mathrm{ORIG}}}$ (E1, largest box at each coupling): 0.383(26) (2.3, 16^4), 0.350(18) (2.3, $16^3 \times 32$), 0.2789(34) (2.4), 0.2072(10) (2.5), against the literature 0.369, 0.266, 0.188 [9, 29]. At $\beta = 2.5$ the E2 ($T = 5$) estimator on the original 16^4 links gives $a^2\sigma = 0.0359(21)$, $a\sqrt{\sigma} = 0.189(6)$, the literature value, so the bias is in E1. In literature units $m_{\mathrm{ORIG}}/\sqrt{\sigma} = 3.54(18)$ (2.4, 16^4) and 3.73(20) (2.5, 20^4). For comparison only, the lowest non-zero-momentum free two-gluon level is $4\sin(\pi/L)$: 1.035 ($L = 12$), 0.780 (16), 0.626 (20); it is not the lowest free-field level in a periodic box (sec. 6).

Table 2. Volume tests (basis A) and ratios at the largest box. Last column: own E1 unit / literature unit.

β	**boxes**	ρ_{VR}	ρ_{ORIG}	ρ_{CP}	**VR call**	$m_{\mathrm{VR}}/m_{\mathrm{ORIG}}$	$m_{\mathrm{VR}}/\sqrt{\sigma}$
2.3	$16^4/12^4$	0.733(111)	1.22(18)	0.981(37)	gapless	0.66(10)	1.80(30) / 1.87(28)
2.3	$16^3 \times 32/12^3 \times 32$	0.847(105)	1.089(67)	1.053(38)	inconclusive	0.673(83)	2.55(33) / 2.42(28)
2.4	$16^4/12^4$	0.984(42)	1.079(62)	1.051(55)	gap	0.711(36)	2.40(8) / 2.52(8)
2.5	$20^4/16^4$	0.944(57)	0.989(72)	0.886(66)	gap	0.639(43)	2.16(10) / 2.38(12)

At $\beta = 2.3$ the VR effective mass falls with t on 8^4, 16^4 and $16^3 \times 32$ and holds only on the 12 boxes; the plateau rule then quotes $m(2)$ on one box and $m(1)$ on the other, which drives the "gapless" call of series 1 and the inconclusive call of series 2. The $\beta = 2.3$ VR channel is not under control. At $\beta = 2.4$ every VR effective mass plateaus; the plateau rule quotes $m(1)$ on both boxes, and with $m(2)$ on both boxes the

volume ratio is 0.96(9) (report only, computed after referee review). At $\beta = 2.5$ $m(1) \approx m(2) \approx m(3)$ in both boxes.

4.3 String tension and the premise check

E1 ratios $\sigma_{\mathrm{VR}}/\sigma_{\mathrm{ORIG}}$: $-0.09(4)$, $-0.04(1)$, $-0.02(1)$ ($\beta = 2.3$; $L = 8, 12, 16$), $-0.048(8)$, $-0.024(10)$ (2.3, $T = 32$), $-0.015(8)$, $+0.016(8)$ (2.4), but 0.219(10) (16^4) and 0.205(7) (20^4) at 2.5: C0 fails at $\beta = 2.5$. CP ratios are 0.94 to 1.17 everywhere (series 1 and 2).

Table 3. Long-time estimators, ratio VR/ORIG, single gauge copy. The $\beta = 2.5$ rows (series 2 configurations) were computed after the C0 failure; the $\beta = 2.4$ rows (series 1 configurations, the stored Wilson loops) were computed after referee review with the same estimator code. At 12^4, $T \leq 6$ leaves no E2 at $T = 6$ and no $\chi(7, 7)$.

β, lattice	E1	E1′ ($R \geq 4$)	E2, $T = 5$	E2, $T = 6$	$\chi(5, 5)$	$\chi(6, 6)$	$\chi(7, 7)$
2.4, 12^4	$-0.015(8)$	0.079(49)	$-0.174(42)$		0.126(20)	$-0.017(45)$	
2.4, 16^4	0.016(8)	0.060(22)	$-0.113(49)$	$-0.083(39)$	0.116(19)	$-0.004(63)$	$-0.017(81)$
2.5, 16^4	0.219(10)	0.191(25)	$-0.079(29)$	$-0.190(62)$	0.429(24)	0.207(56)	0.062(101)
2.5, 20^4	0.205(7)	0.213(13)	$-0.046(21)$	$-0.029(52)$	0.397(19)	0.237(50)	0.044(80)

The VR Creutz ratios (relative to ORIG) at $\beta = 2.5$, 16^4 fall $0.97, 0.83, 0.63, 0.43, 0.21, 0.06$ for $R = 2$ to 7 ; at $\beta = 2.4$, 16^4 they fall $0.86, 0.56, 0.28, 0.12, 0.00, -0.02$; at $\beta = 2.3$ ($16^3 \times 32$) they are $0.68, 0.27, 0.07$ for $R = 2$ to 4 and zero within errors beyond. The VR potential rises at short distance and is flat within errors beyond $R \approx 4a$ (2.3), $6a$ (2.4) and $7a$ (2.5), roughly 1.4 to $1.7/\sqrt{\sigma}$ at each coupling. E1 fits $T = 3 \to 4$ over $R = 2$ to $L/2$ with a free Coulomb term, so it is dominated by the rising intermediate-distance part when the flattening lies well inside the window (2.5), while at 2.4 the free e absorbs part of the rise and E1 reads zero. The negative E2 values are a fit artifact of $V_0 + \sigma R - e/R$ with free e on a potential that rises and then flattens, not a negative string tension; the same functional form fails on VR potentials in [16].

What this means for the $\beta = 2.4$ premise (analysis done after referee review, not pre-registered, reported because it bears on the headline coupling): every long-time ratio at 2.4 is below the pre-registered threshold 0.15, and the VR string tension is consistent with zero at the largest distances we resolve ($\chi(6, 6)$, $\chi(7, 7)$, $R = 6, 7$, about 1.6 to $2.0/\sqrt{\sigma}$). It is not zero at intermediate distances ($\chi(5, 5) = 0.116(19)$, E1 $' = 0.060(22)$). So the statement “the VR string tension is zero within errors” at 2.4 holds for E1 and at long range, not at all distances; the 2.4 and 2.5 ensembles differ mainly in where the flattening falls relative to the E1 window. Item (b) of sec. 2 is this quantification.

4.4 Gauge-copy study (series 3)

Rules were fixed before the series 3 data existed. The best-of-16 functional exceeds the single copy on 94% of the 668 16^4 configurations at $\beta = 2.5$, by 0.00215(5) on average (0.79115 → 0.79330). For comparison, [14] reports a mean maximal functional of about 0.797 to 0.798 on the same lattice (16^4,

$\beta = 2.5$) with simulated annealing and many copies, so our best-of-16 copies are not close to the global maximum. The single-copy E1 ratio 0.205(9) reproduces series 2 (0.219(10)) on independent configurations, and the $\beta = 2.3$ control passes ($-0.046(21)$).

Table 4. Series 3, 16^4, $\beta = 2.5$, $N = 668$: string-tension ratios to ORIG, single copy vs best of 16 (paired difference).

estimator	VR single	VR best of 16	VR diff.	CP single	CP best of 16	CP diff.
E1	0.205(9)	0.225(10)	+0.020(13)	0.985(19)	0.726(14)	−0.259(12)
E2, $T = 5$	−0.042(37)	−0.044(37)	−0.002(50)	1.086(70)	0.765(50)	−0.321(28)
E2, $T = 6$	−0.045(95)	−0.151(109)	−0.106(135)	1.70(33)	1.16(22)	−0.54(11)
$\chi(6,6)$	0.232(72)	0.357(66)	+0.125(83)	1.013(94)	0.728(67)	−0.285(31)
$\chi(7,7)$	0.188(217)	0.106(200)	−0.082(285)	1.35(51)	0.93(35)	−0.42(16)

Ranking the 16 candidates by functional, the CP E1 ratio falls almost monotonically from 1.32 (worst) to 0.73 (best), while the VR E1 ratio rises mildly from 0.17 to 0.23. At $\beta = 2.5$, 16^4, the VR glueball mass (basis A) moves from 0.451(22) (plateau) with the single copy to 0.561(44) (late) with the best copy, a paired difference of $+0.109(48)$, 2.3σ ($m(1)$: $+0.051(28)$); the single-copy value agrees with series 2 (0.475(18)). At the $\beta = 2.3$ control ($12^3 \times 32$, 241 configurations) the VR mass moves the other way, from 1.152(106) to 0.961(60) (both plateau), a paired difference of $-0.192(109)$, -1.8σ. The 20^4 sample at $\beta = 2.5$ (54 configurations) gives E1 VR ratios 0.223(17) (single) and 0.252(23) (best), CP 0.970(22) and 0.754(22). It is too small for masses: its single-copy VR mass comes out 0.300(74) ("late"), against 0.448(21) in series 2.

5. Pre-registered outcomes

Each series fixed its combined rule in advance; we report them as written.

- Series 1: "gapless", triggered by $\beta = 2.3$ alone ($\rho_{\mathrm{VR}} = 0.733(111)$) while $\beta = 2.4$ was a "gap".
- Series 2: the combined rule as written reads "gap survives vortex removal (in this window)" ($\beta = 2.5$ gap, 2.4 gap, no significant drop of $m_{\mathrm{VR}}/\sqrt{\sigma}$ from 2.4 to 2.5, 2.3 not gapless). The rule did not say whether the series 1 consequence of a C0 failure (that coupling does not enter the verdict) carries over; with it, the outcome is inconclusive.
- Series 3 was designed to settle that by test. Its rule: if the best-of-16 VR E1 ratio is still ≥ 0.15, the combined outcome is inconclusive, because the $\beta = 2.5$ VR point is not a clean vortex-free measurement under C0. The ratio is 0.225(10), so the **combined outcome is inconclusive**. The pre-registered gauge-fixing diagnosis (G) does not hold: with best-of-16 copies the residual does not fall (it rises slightly with the functional). Copies closer to the global maximum (simulated annealing, [14]) and Laplacian center gauge [2] were not tested. "Measurement artifact of the short-time estimator" and "real residual" are not separated by the pre-registered test, because the long-range estimators at 16^4 have errors of about 0.2.

6. Discussion

The cleanest point in this data set is $\beta = 2.4$. There the vortex-removed 0^{++} channel has a decay rate of about 0.7 of the original glueball mass that does not change between 12^4 and 16^4: $\rho = 0.984(42)$, 5.6σ from the ratio 0.75 that a level scaling as $1/L$ would give (0.96(9) with $m(2)$). This volume ratio is the evidence against a finite-volume state, and it has two limits. First, the free two-gluon level we compare with, $4\sin(\pi/L)$, is the lowest non-zero-momentum level; in a periodic box the lowest free-field 0^{++} levels lie in the zero-momentum (torus) sector, with energies of order $g^{2/3}/L$ at weak coupling [18, 19], which can be below it. Being below $4\sin(\pi/L)$ therefore does not by itself exclude a finite-volume state, and we keep that comparison as secondary only. Second, with $\sigma_{\mathrm{VR}} = 0$ at long range a pair of Polyakov loops (flux winding the box) has no energy cost linear in L, so a volume-independent level could also be such a finite-volume state; we did not measure Polyakov loops. The $\beta = 2.4$ point rests on two couplings for any comparison, one gauge copy (with an untested copy systematic that was about 20% of the mass at the neighbouring couplings), and a string tension that is zero within errors with E1 and at the largest distances, but not at intermediate distances (sec. 4.3). Its mass is $2.40(8)\sqrt{\sigma}$ in our own unit and $2.52(8)\sqrt{\sigma}$ in the literature unit; the ratio 0.71(4) to the original mass does not depend on the unit.

At $\beta = 2.5$ the same picture holds for the mass, and the long-time estimators are consistent with [1, 2] at the largest distances we can resolve ($R \approx 7a$, about 1.3 to $1.5/\sqrt{\sigma}$), but the pre-registered short-time check does not pass, and the VR mass moves at the 2σ level with the gauge copy. The center-projected string tension depends strongly on the gauge copy (0.99 of the full one with a single copy, 0.73 with the best of 16, falling almost monotonically with the functional, as in [14, 15]); since the VR links are the original links with the center part divided out, the VR ensemble itself depends on the copy, and a VR mass should be quoted with that systematic. At $\beta = 2.3$ the VR correlator is not a clean exponential at the available times.

What would settle it: a premise check built on long-time or Creutz-ratio estimators fixed before new data, larger T and finer couplings, a third box per coupling, Polyakov-loop measurements on the VR ensembles, and a systematic gauge-copy study of the VR mass at every coupling (including simulated annealing and Laplacian center gauge).

Data and code

Data tables (CSV) and the simulation and analysis code for all three series accompany this note in the same repository. Run on the original measurement files, the series 1 and series 2 analysis scripts reproduce data/series1_summary.csv and data/series2_summary.csv exactly. The measurement files (several GB) are not included; the code regenerates statistically equivalent ensembles from the listed seeds.

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Entries marked [recalled] were not opened in full text for this note. The bibliographic data of every entry were checked against arXiv or INSPIRE metadata.

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